Study & Demonstration of GlassWire: Network Monitoring and Security Analysis

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1 ABSTRACT

GlassWire is a graphical network monitoring and security analysis tool that helps users observe real-time internet activity, application-level bandwidth usage, remote hosts contacted, country-wise traffic distribution, and connected devices on the local network. Unlike traditional vulnerability scanners, GlassWire focuses on visibility, monitoring, and alert-based security awareness.

In this experiment, GlassWire was installed and demonstrated on a Windows system. Different features such as Traffic Monitor, Log Analysis, GlassWire Protect (Firewall), Network Scanner, and Hardware Resources were explored. During execution, the tool successfully identified first-time network activity from multiple applications, displayed high bandwidth usage by Google Chrome, mapped network traffic to Google Video/YouTube-related hosts, showed country-wise traffic distribution, and detected multiple devices connected to the same Wi-Fi network.

The results confirm that GlassWire is an effective tool for network monitoring, application behavior analysis, local network visibility, and basic endpoint security awareness. Although the free version has limited firewall controls, it remains a highly useful tool for educational demonstrations and security analysis.

2 OBJECTIVES

The objectives of this experiment are:

- To study the working of GlassWire as a network monitoring and security analysis tool
- To monitor real-time network traffic generated by applications
- To identify which applications consume the most bandwidth
- To inspect remote hosts/IP addresses contacted by the system
- To observe alerts for first-time network activity
- To understand the GlassWire Protect / Firewall view and connection states
- To detect devices connected to the local network using Network Scanner
- To monitor CPU, Memory, Disk, and GPU usage through Hardware Resources
- To evaluate the usefulness of GlassWire in security visibility and monitoring

3 INTRODUCTION

In today's digital environment, applications continuously communicate with remote servers for updates, synchronization, cloud access, advertisements, and content delivery. Without proper monitoring, users may remain unaware of which applications are using the internet, what remote servers they are contacting, or whether suspicious activity is taking place in the background.

GlassWire is a network monitoring and security analysis tool designed to solve this problem by providing a clear graphical view of network activity. It allows users to monitor:

- Live upload and download traffic
- Application-wise network usage
- Remote hosts contacted
- Country-wise traffic flow
- Alerts for new or first-time network activity
- Connected devices on the local network
- Hardware resource usage

This makes GlassWire useful for:

- Detecting unusual network behavior
- Identifying high bandwidth-consuming applications
- Observing unknown background activity
- Recognizing new devices on the network
- Improving overall security awareness

Thus, GlassWire is not just a monitoring tool but also a security visibility tool that supports system, network, and application-level analysis.

4 FEATURES OF THE TOOL

The following features of GlassWire were explored during the experiment:

4.1 Traffic Monitor

The Traffic Monitor provides real-time visibility into network activity. It allows analysis of:

- Applications using the internet
- Hosts / domains contacted
- Traffic types such as HTTPS, DNS, mDNS, etc.
- Countries involved in the communication
- Bandwidth spikes over time through graphical visualization

This feature is useful for identifying which applications are actively consuming data and what servers they are communicating with.

4.2 Log Analysis

The Log Analysis tab records security-relevant events such as:

- First-time network activity by an application
- New network connections
- Background application communication

This helps detect when an application first accesses the internet and provides visibility into the remote domain or IP involved.

4.3 GlassWire Protect / Firewall

The GlassWire Protect section shows:

- Active applications
- Allowed / blocked connection states
- Inbound and outbound visibility
- Hosts contacted
- Limited firewall control in the free version

Although advanced controls are restricted in the free edition, it still provides useful application-level connection visibility.

4.4 Network Scanner

The Network Scanner detects devices connected to the same local network and displays:

- Device name
- Device type
- Operating system
- IP address
- MAC address
- First seen / last seen timestamps

This is helpful for identifying unknown or unauthorized devices on a Wi-Fi network.

4.5 Hardware Resources

The Hardware Resources tab monitors:

- CPU usage
- Memory usage
- Disk transfer rate
- GPU usage
- GPU memory

This helps correlate network activity with system resource consumption, which can be useful during security analysis.

5 INSTALLATION & SETUP

The following steps were performed to install and set up GlassWire:

Step 1: Download

GlassWire was downloaded from the official website.

Step 2: Install

- The installer was executed on a Windows system.
- Standard installation steps were followed.

Step 3: Launch

- After installation, GlassWire was launched successfully.

- The main dashboard displayed the following tabs: Traffic Monitor, GlassWire Protect, Log Analysis, Network Scanner, Hardware Resources.

Step 4: Generate Network Activity
To demonstrate the tool, normal internet activity was generated by:

- Opening Google Chrome
- Visiting websites
- Streaming media / online video
- Allowing background applications to communicate

This generated measurable traffic that could be analyzed in the tool.

6 EXPERIMENTATION & RESULTS

During the experiment, multiple GlassWire features were demonstrated and analyzed. The following results were observed.

6.1 Log Analysis – First Network Activity Detection

The Log Analysis tab reported several applications initiating their first network connection. Applications observed included:

- Microsoft Office SDX Helper
- Adobe Acrobat
- Brave Browser
- MicrosoftWindows.Client.WebExperience
- MoUSO Core Worker Process
- LenovoVantage-(GenericMessagingAddin).exe
- OneDriveLauncher
- ActionsServer.exe

This indicates that GlassWire can detect when an application first accesses the network and report the corresponding destination host.
This screenshot shows the Log Analysis tab where GlassWire records first-time network activity by several applications. Each entry shows the application name, the remote host contacted, and the timestamp. This is useful for detecting unexpected or newly initiated network communication.

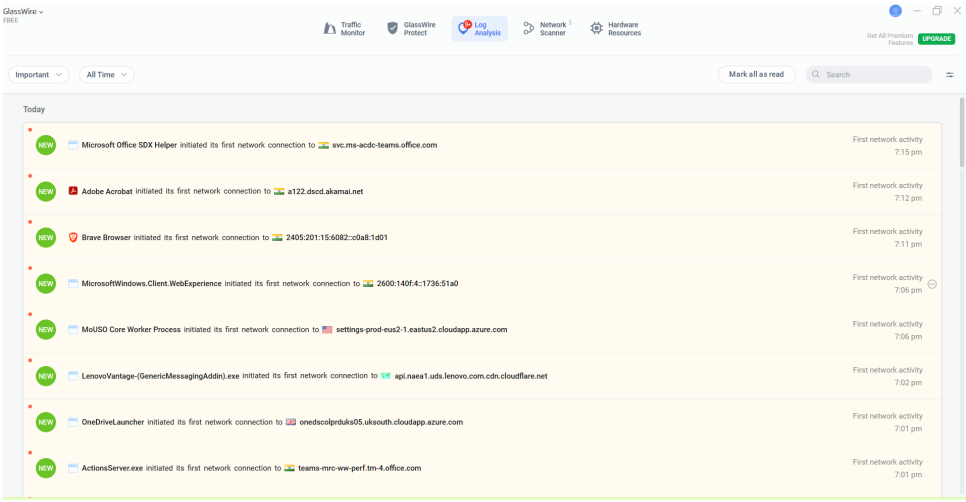

Figure 1: Log Analysis showing first network activity alerts for multiple applications.

6.2 Real-Time Alert Popup

A real-time alert popup was generated for Adobe Acrobat, indicating its first network connection to:

- `a122.dscd.akamai.net`

This demonstrates that GlassWire provides instant notifications when an application first uses the network.

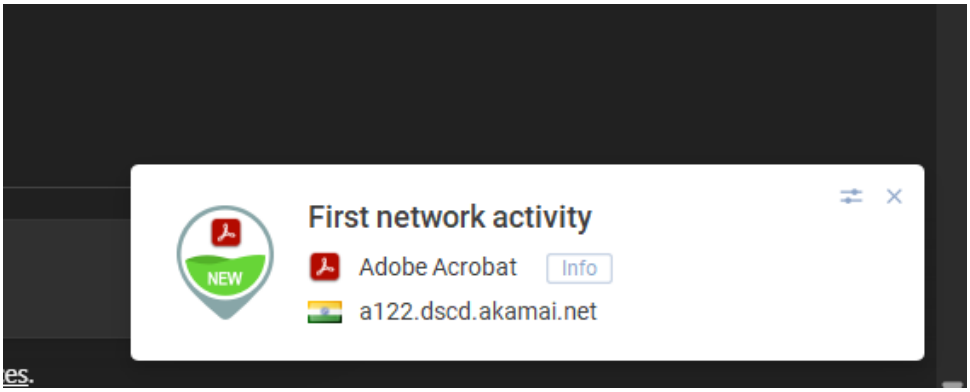

Figure 2: Real-time alert popup for first network activity by Adobe Acrobat.

This screenshot shows a real-time GlassWire alert popup indicating that Adobe Acrobat initiated its first network connection. The remote host displayed is `a122.dscd.akamai.net`. This helps in identifying new or unexpected network behavior immediately.

6.3 Traffic Monitor – Real-Time Network Usage

The Traffic Monitor displayed real-time network usage in multiple categories: Applications, Hosts, Traffic Types, Countries. During the demonstration:

- Google Chrome was the most active application

- Multiple Google Video / YouTube-related hosts were contacted
- Traffic was distributed across several countries

This confirms that the tool can accurately map network activity to applications and destinations.

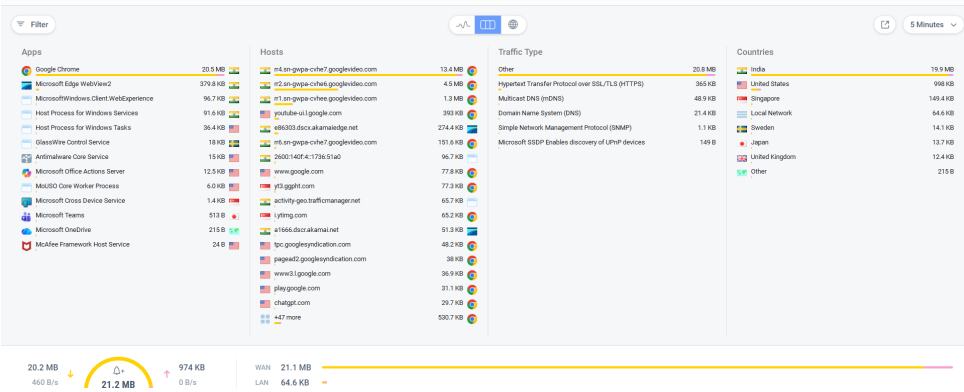

Figure 3: Traffic Monitor showing app-wise, host-wise, traffic type, and country-wise network usage.

This screenshot shows the Traffic Monitor in table view. It breaks down traffic by application, remote host, protocol type, and country. The data confirms that GlassWire can visually represent network activity in a structured and security-relevant format.

6.4 Host-Level Inspection

A selected host popup displayed details of the remote server:

- **Host:** rr4.sn-gwpa-cvhe7.googlevideo.com
- **Region:** India
- **IP:** IPv6 address
- **Associated Application:** Google Chrome

This demonstrates GlassWire's ability to map a specific remote host to a specific application.
This screenshot shows the detailed view of a selected remote host. It identifies the host, its region, IP address, and the application responsible for the connection. This is useful for investigating unknown or suspicious destinations.

6.5 Graphical Traffic Visualization

The graphical view displayed traffic spikes during periods of internet activity. These spikes indicated increased upload/download bandwidth usage. This proves that GlassWire provides real-time graph-based visualization of network activity.

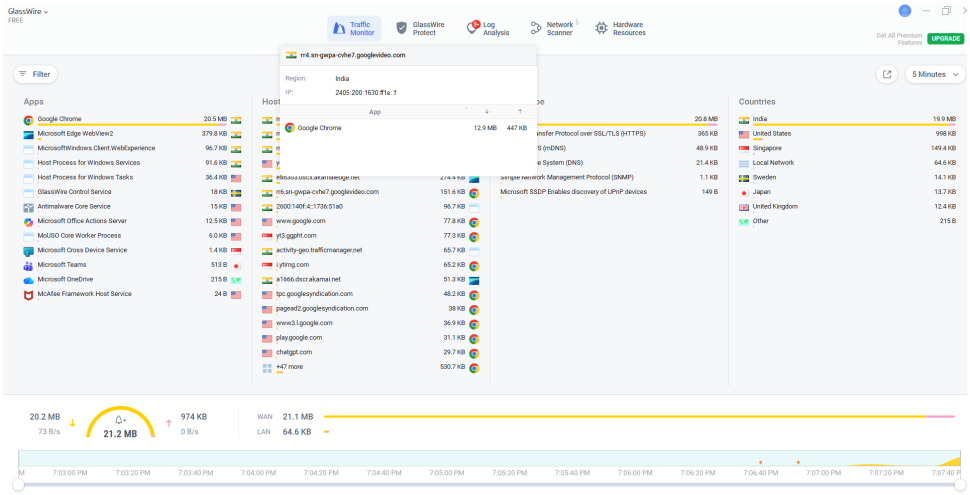

Figure 4: Host detail popup showing remote host, region, IP, and associated application.

Figure 5: Traffic graph showing live network activity and bandwidth spikes over time.

This screenshot shows the graphical traffic view in GlassWire. The visible peaks represent increased network activity during the experiment. This feature is useful for quickly identifying sudden bursts of traffic or abnormal background usage.

6.6 Detailed Traffic Session Analysis

In a higher-usage traffic session, the following values were observed:

- **Application Usage:** Google Chrome (103.7 MB)
- **Top Host:** rr2.sn-gwpa-cvhy.googlevideo.com (94.7 MB)
- **Traffic Types:** Other (104.3 MB), HTTPS (7.3 MB), DNS (54.6 KB), mDNS (141 KB)
- **Country Distribution:** India (99.7 MB), Other (5.1 MB), United States (4.2

MB), Singapore (1.2 MB). Additional minor traffic to Japan, Local Network, Australia, Korea, UK, and Sweden.

These results strongly indicate browser-based media/video traffic, likely due to streaming or content delivery.

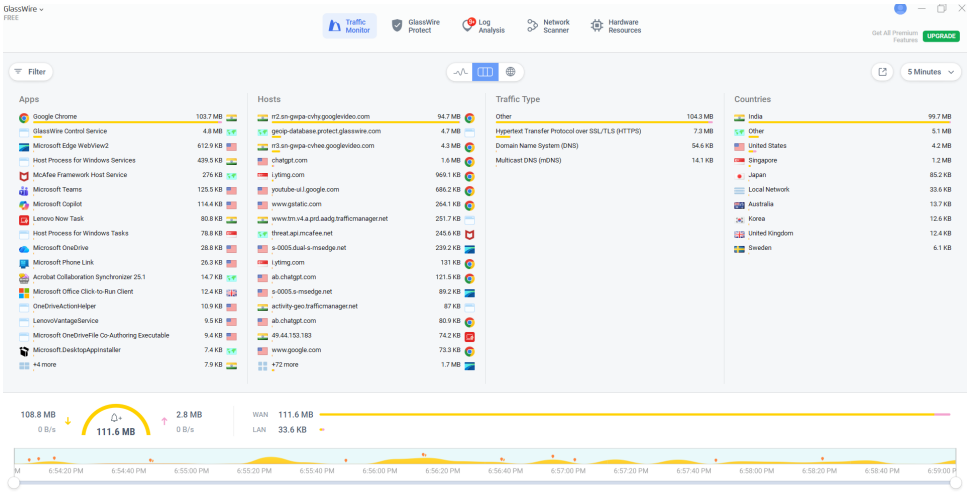

Figure 6: Detailed Traffic Monitor session showing high network usage by Google Chrome.

This screenshot shows a high-traffic session where Google Chrome consumed 103.7 MB of data. The dominant host is rr2.sn-gwpa-cvhy.googlevideo.com with 94.7 MB, indicating media or video content delivery. The traffic type and country breakdown provide deeper insight into the nature of the communication.

6.7 GlassWire Protect / Firewall

The GlassWire Protect tab displayed:

- Application names
- Inbound and outbound connection states
- Allowed / blocked indicators
- Host information
- Traffic values

In the free version, advanced blocking options are limited, but the interface still provides important connection visibility.
This screenshot shows the GlassWire Protect tab, which lists active applications and their connection states. It helps identify which applications are currently communicating over the network and whether those connections are allowed or blocked.

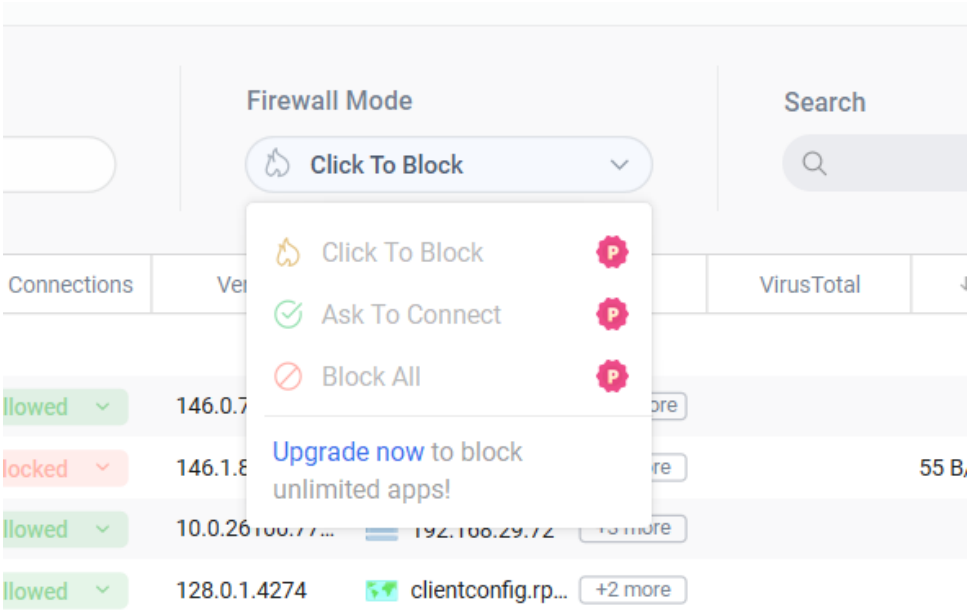

Figure 7: GlassWire Protect / Firewall tab showing application connection states.

6.8 Firewall Mode Limitation (Free Version)

The firewall mode menu displayed:

- Click To Block
- Ask To Connect (Premium)
- Block All (Premium)

This confirms that the free version offers limited firewall control and that some advanced features require an upgrade.
This screenshot shows the firewall mode selection menu. It indicates that only basic options are available in the free version, while advanced modes such as Ask To Connect and Block All require the premium edition.

6.9 Network Scanner – Local Device Detection

The Network Scanner successfully detected devices connected to the SRK_5G network, including:

- Akshat – Lenovo – Windows
- Android – Vantiva USA – Android
- RSVOTCGM0020607 – Reliance router/device
- Mobile Phone – Apple iPhone – iOS
- Generic device

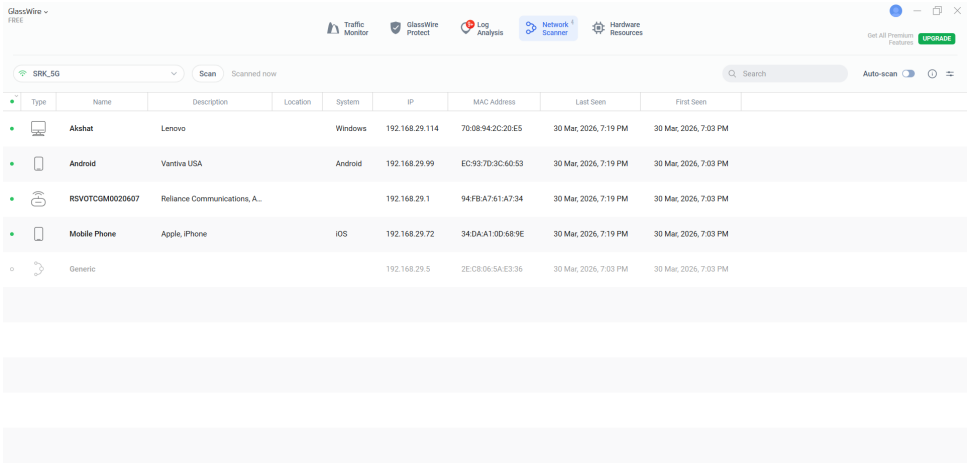

Figure 8: Network Scanner showing devices connected to the local Wi-Fi network.

The tool displayed IP addresses, MAC addresses, and first seen/last seen timestamps. This screenshot shows the Network Scanner tab where GlassWire detects devices on the local network. It provides device identification, IP addresses, MAC addresses, and timestamps, which can help identify unauthorized or unknown devices connected to the Wi-Fi.

6.10 Hardware Resources Monitoring

The Hardware Resources tab showed CPU usage over time, Memory usage, Disk transfer rate, GPU usage, and GPU memory. This feature helps correlate network activity with system performance.

Figure 9: Hardware Resources tab showing CPU, Memory, Disk, and GPU usage over time.

This screenshot shows the Hardware Resources tab in GlassWire. It monitors CPU, Memory, Disk transfer rate, and GPU usage. This can be useful in security analysis when unusual network activity also causes abnormal system resource usage.

7 ANALYSIS OF OUTCOME

Based on the experiment, the following outcomes were analyzed:

1. **Successful Detection of First-Time Network Activity**
   GlassWire successfully identified when multiple applications initiated their first network connection. This is useful for identifying unexpected or newly active applications.

2. **Effective Real-Time Traffic Monitoring**
   The Traffic Monitor clearly displayed network usage in real time. It showed which applications consumed bandwidth, which remote hosts were contacted, what protocols were used, and which countries were involved.

3. **Clear Identification of High Bandwidth Consumption**
   Google Chrome was consistently observed as the highest bandwidth-consuming application. In one session, it used 103.7 MB, with most traffic directed to googlevideo.com hosts, indicating video/media streaming.

4. **Host-Level Visibility Improves Security Awareness**
   The ability to inspect remote hosts and link them to specific applications helps users determine whether a connection is expected or suspicious.

5. **Local Network Awareness**
   The Network Scanner detected all visible devices connected to the SRK_5G network, demonstrating that GlassWire can improve awareness of local network devices and help identify unknown systems.

6. **Firewall Visibility is Useful but Limited in Free Version**
   The GlassWire Protect section provided useful visibility into application connection states, but the free version limited advanced blocking options.

7. **Hardware Monitoring Adds Context**
   The Hardware Resources tab added system-level visibility, making it easier to correlate network activity with CPU, memory, disk, and GPU behavior.

Overall Outcome:
The experiment proved that GlassWire is a highly effective tool for network monitoring, application behavior analysis, alert-based security awareness, local network visibility, and basic endpoint security monitoring.

8 CONCLUSION

The experiment successfully demonstrated GlassWire as a network monitoring and security analysis tool. The tool was able to provide clear and detailed visibility into system network behavior, including:

- Real-time traffic monitoring
- Application-wise data usage
- Host and country-level traffic analysis
- Alerts for first-time network activity

- Firewall-related connection visibility
- Local network device detection
- Hardware resource monitoring

The results showed that Google Chrome generated the highest traffic, especially during browsing or media-related activity, and most high-volume traffic was linked to Google Video / YouTube-related hosts. GlassWire also successfully detected multiple applications initiating first-time network connections and identified several devices connected to the local Wi-Fi network.

Although the free version has some limitations in firewall controls, GlassWire remains an excellent tool for:

- Security monitoring
- Application behavior observation
- Network awareness
- Educational demonstrations

Therefore, GlassWire is a suitable and effective tool for this internal assessment under the category of network/system/application security monitoring and analysis.